Zhihong Zhang

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Industrial Data Analysis × Predictive Maintenance — PHM · IoT Pipelines · Ops Dashboards

Summary

Industrial data analyst with a chemical engineering and mathematics background. At Sinopec Hydrogen Energy I built multi-sensor IoT pipelines and equipment-health dashboards (failure rate down ~35%) and used daily/weekly operations data to tune compressor scheduling (energy per kg H2 down ~12%); led joint-university compressor predictive-maintenance research (TensorFlow with failure-mode knowledge, scarce samples, low SNR). Daily analysis in Python/pandas/SQL with Tableau dashboards; AI tools like Claude Code speed up my script work. Bilingual; 8 years in the US.

Experience

Sinopec Hydrogen Energy (Shanghai) Co., Ltd. — Shanghai, China
IT Operations & Data Analyst — Jan 2023 – Jun 2026

- On-site at hydrogen refueling stations: commissioned multi-sensor systems and industrial equipment (compressors, dispensers); explained model outputs to operations teams so maintenance decisions relied on data, not guesswork.
- Technical lead for joint-university research on diaphragm compressor predictive maintenance: combined TensorFlow deep-learning models with compressor failure-mode knowledge under scarce-sample, low-SNR conditions.
- Built real-time equipment-health dashboards that turned sensor time-series into signals the shift could act on (failure rate down ~35%); analyzed daily/weekly operations data to tune compressor scheduling and pressurization (energy per kg H2 down ~12%).
- Used Claude Code and own multi-agent workflows to automate analysis-script writing and validation; routine script work ~3x faster.

Freelance / Independent Consulting — Shanghai, China
Data Analysis & Financial Modeling — Dec 2021 – Dec 2022

- Did data analysis for 2-3 early-stage startups — Python (pandas, scikit-learn) for processing, feature analysis and modeling; delivered visual reports clients used directly.
- Data analysis services for multiple clients: stock signal research, backtesting frameworks, financial time-series analysis.
- Built automated data pipelines and dashboards, turning portfolio risk assessment and market-trend analysis from manual work into automation.

Yeshiva University — New York, USA
Data Science Graduate Project — Aug 2019 – Jan 2021

- Capstone: tuned a PyTorch CNN for speaker verification (EER down 10%); insurance risk prediction on imbalanced data (scikit-learn/TensorFlow, 73% to 88%).

Selected Projects

- **H2 Station Anomaly Early-Warning System** **[response time ↓80%]** — multi-sensor real-time pipeline, anomaly detection & tiered alerting *(Proprietary; architecture and alerting-strategy walkthrough available in interview)*
- **Compressor Predictive Maintenance** **[failure rate ↓35%]** — fault prediction under scarce-sample, low-SNR conditions: mechanism-informed features × TensorFlow
- **A-share Data Toolkit (open-source): market dashboards + heatmap auto-analysis** — Two open-source tools: StockCN Analyst Toolkit ingests indices/sectors/stocks via zero-key public APIs (Tencent, East Money, Sina) into interactive HTML dashboards; a-share-heatmap-analysis scrapes the market heatmap and auto-generates analysis reports. CLI-first, consumable by AI agents (GitHub: steinszzh)
- **COVID-19 × Stock Market × Unemployment Data Warehouse (YU Course Project)** **[AWS RDS + Tableau]** — Python (pandas) ETL of multi-source data (NYT, FRED, DoL) into an AWS RDS MySQL instance: staging tables -> dimensional model -> fact tables, with Tableau dashboards (Yeshiva course project)
- **Agent Orchestra — Multi-Agent Orchestration Framework** **[49 tests passing]** — Open-source, zero-dependency multi-agent orchestration framework: 6 patterns (sequential/parallel/router/hierarchical/debate/DAG), LLM backend abstraction & capability routing

Skills

Engineering & Data	Python, SQL, MATLAB, TensorFlow, scikit-learn, Tableau; AI-assisted dev tools (Claude Code, etc.) for daily analysis work
Time-Series	Anomaly detection, predictive-maintenance modeling, feature engineering, denoising, imbalanced data (grounded in production industrial projects)
Industrial & Cloud	IoT multi-source sensor pipelines, edge-to-cloud streaming, Linux, Docker
Domain Knowledge	Chemical process & equipment mechanisms (compressors, H2 stations), failure-mode analysis, Bayesian statistical modeling, PID control systems
LLM & AI Agents	Local LLMs (llama.cpp/Ollama), RAG & vector search, multi-agent orchestration (agent-orchestra, open-source)

Education

Yeshiva University (GPA 3.9/4.0) — M.S. Data Analytics and Visualization — 2019 – 2021, New York, USA

Calvin University — B.S. Mathematics (Minors: Engineering, Chemistry) — 2012 – 2017, Michigan, USA

Technical University of Berlin (TU Berlin) — Summer School — Civil Engineering Statics & Dynamics, German — Jun 2013 – Aug 2013, Berlin, Germany